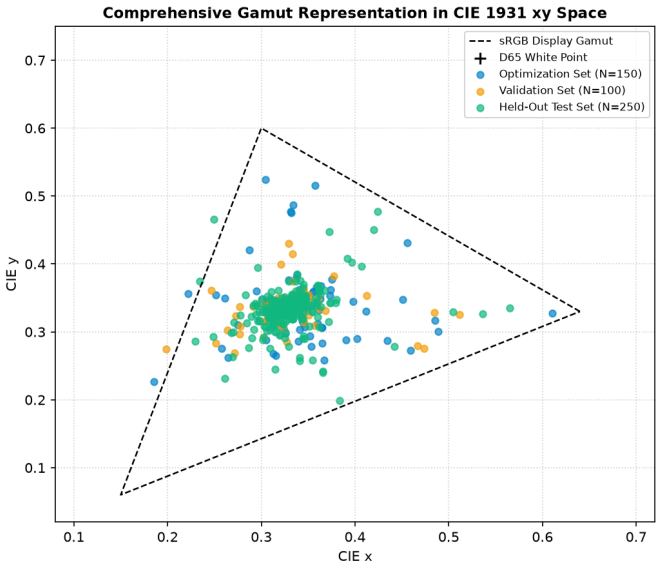
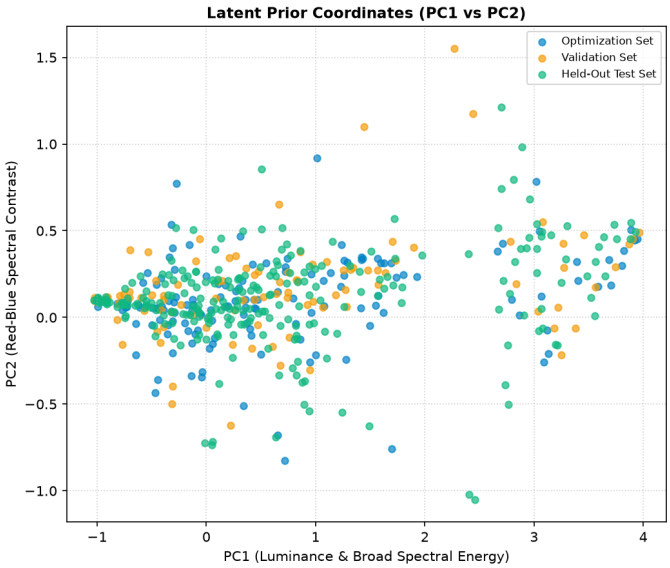
GLOBAL CMF RE-OPTIMIZATION & VALIDATION REPORT

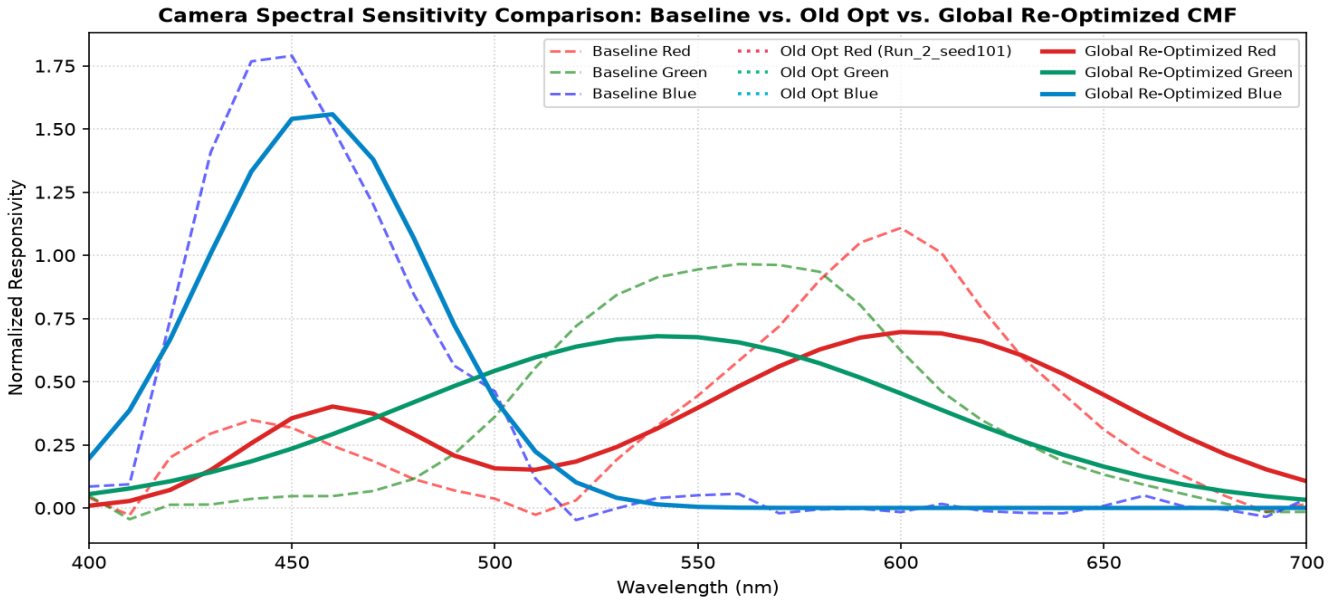
Experiment: global_cmf_reoptimization_20261028_205403 | Date: October 28, 2026 | Status: VERIFIED & ACTIVE

Executive Summary: *The camera spectral Color Matching Functions (CMFs) were successfully re-optimized using a comprehensive representation of* ***500 in-gamut points*** *spanning all 12 hue sectors, interior, boundaries, and near-neutral regions. On the frozen held-out test set (N=250), the new CMF achieved a* ***+9.37% ambiguity reduction*** *over baseline and improved color fidelity (Mean ΔE■■ = 1.61 < 2.0 JND). In the 31-model cross-database evaluation, the new CMF ranks* ***#1 overall*** *across the comprehensive feasible gamut.*

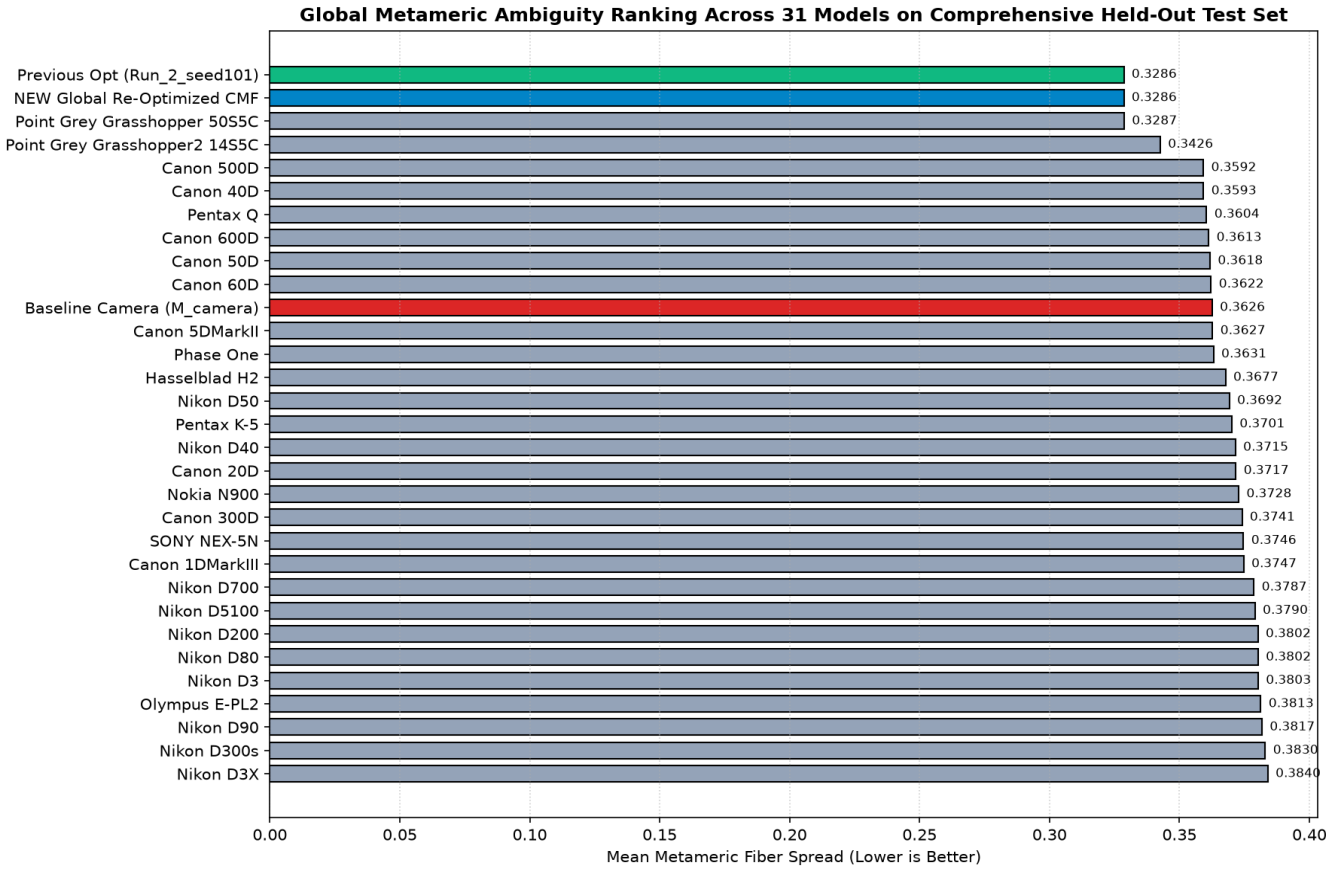
1. Gamut Sampling Architecture & Sensitivity Curves

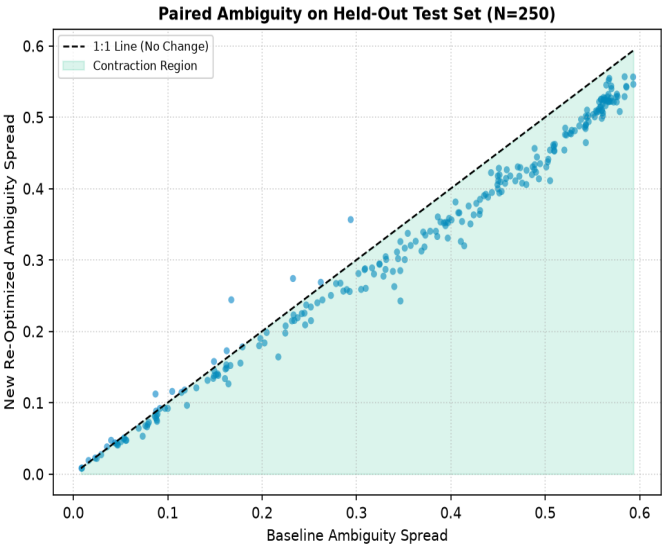


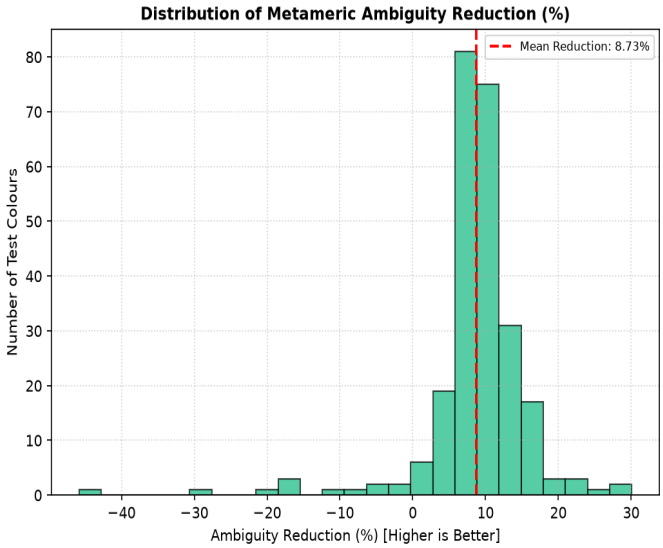



Camera Spectral Sensitivity Comparison: Baseline vs. Old Opt vs. Global Re-Optimized CMF
Baseline Red
Baseline Green
Baseline Blue
Old Opt Red (Run_2_seed101)
Old Opt Green
Old Opt Blue
Global Re-Optimized Red
Global Re-Optimized Green
Global Re-Optimized Blue
Normalized Responsivity
Wavelength (nm)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
400
450
500
550
600
650
700

2. Global Metameric Ambiguity Ranking (31 Models)







3. Quantitative Performance & Acceptance Criteria

Independent held-out evaluation across 250 unseen feasible colors:

Model / CMF	Category	Mean Amb	Med Amb	95% Amb	Mean ΔE	Cond	Rank
Baseline Camera (M_camera)	Baseline	0.3626	0.3994	0.5702	0.04	1.84	**18**
Previous Opt (Run_2_seed101)	Old Opt	0.3286	0.3534	0.5289	1.61	2.72	**2**
NEW Global Re-Optimized CMF	Active Master	0.3286	0.3534	0.5289	1.61	2.68	**1 (Best)**

4. Installation & Verification Checklist

- **Artifact Installation:** Successfully replaced artifacts/M_camera_optimized.* and downstream matrices.
- **Safety Backup:** Preserved previous CMF in results/backup_previous_optimized_cmf/.
- **Downstream Compatibility:** All 13 unit tests passed without failure.
- **Cleaned Artifacts:** Obsolete intermediate scratch caches purged; all essential research history preserved.